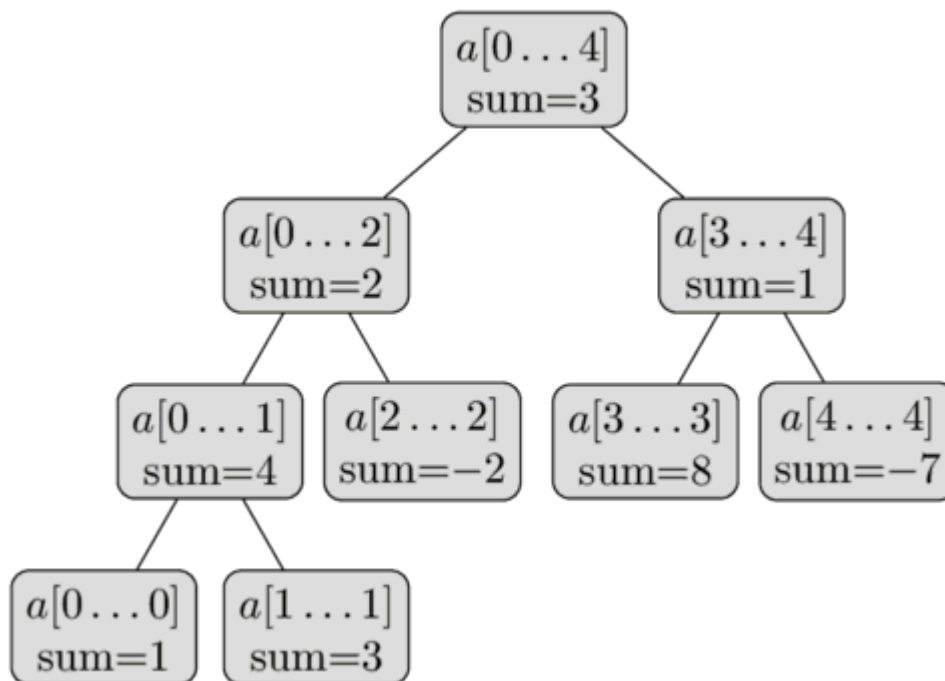


Segment Tree:

Segment tree is a tree based(binary tree) data structure used for storing information about intervals or segments. It allows answering range queries over an array efficiently .This includes finding sum of a range or finding minimum of a range in $O(\log_2(n))$ time complexity. Tree allows modifying the array by replacing one element, or even changing the elements of a whole subsegment (e.g. assigning all elements $a[l \dots r]$ to any value, or adding a value to all element in the subsegment).It follow the strategy of divide and conquer .Suppose we want to calculate the cumulative sum of the array. Then first we divide the array into two parts then recursively divide them until we have only one element left. These final elements will be the leaf nodes of the binary tree and we can use directly their value for calculating the value for their ancestor. Generally ancestor calculate his value using the sum of his left and right child.

$A[] = \{1, 3, -2, 8, -7\}$



For applying this algorithm in programmatically we use linear array. Generally for an n size array the size of the segment tree will be at max $4n$. In general in segment tree if an index has value the value of left child will be at $2 * \text{index}$ and right child will be at $2 * \text{index} + 1$. So first element of array will be the result of the whole array.